



Machine learning Algorithms

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Machine learning Algorithms



- I. Introduction to machine learning
- II. Supervised Machine Learning Algorithms
 - I. Decision Trees
 - II. Sequential covering algorithm
 - III. K-Nearest Neighbors (KNN)
 - IV. Support vector machines
- III. Ensemble learning: bagging, Boosting
- IV. Unsupervised machine learning Algorithms
 - I. FP-Growth algorithm
 - II. Mean-Shift algorithm
 - III. Expectation–Maximization (EM) Clustering using Gaussian Mixture Models (GMM)
 - IV. OPTICS, MDCUT
 - V. Divisive hierarchical clustering
 - VI. Affinity propagation
 - VII. BIRCH
- V. Introduction to Reinforcement Learning

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— Chapter 1 — Introduction to Machine Learning

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Machine Learning



- Machine learning is a branch of artificial intelligence (AI) and computer science
- Machine learning is an important component of the growing field of data science.
- Learning: Field of study that gives computers the ability to learn without being explicitly programmed. Arthur Samuel's checkers player (1959).
- Looking for patterns in vast amounts of data that is just too big for humans to analyze

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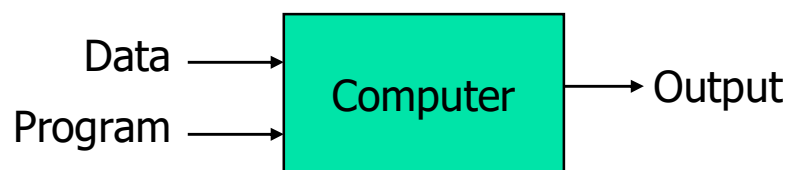
What is Learning?

- "Learning denotes changes in a system that ... enable a system to do the same task ... more efficiently the next time." - Herbert Simon
- "Learning is constructing or modifying representations of what is being experienced." - Ryszard Michalski
- "Learning is making useful changes in our minds." - Marvin Minsky

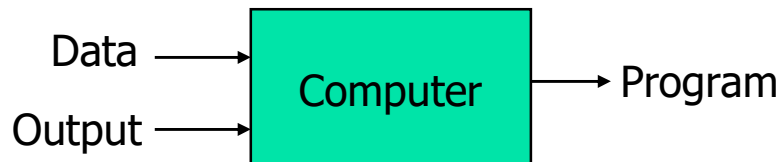
"Machine learning refers to a system capable of the autonomous acquisition and integration of knowledge."

Machine Learning

Traditional Programming



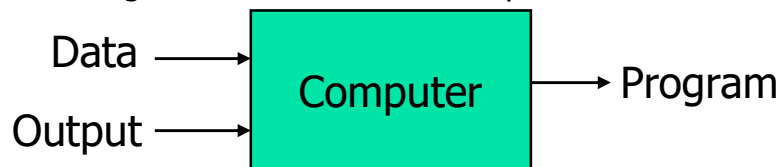
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Types of Learning

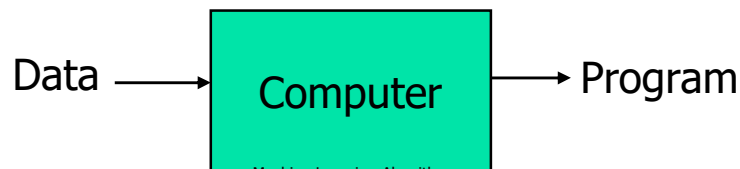
■ Supervised (inductive) learning

- Training data includes desired outputs



■ Unsupervised learning

- Training data does not include desired outputs



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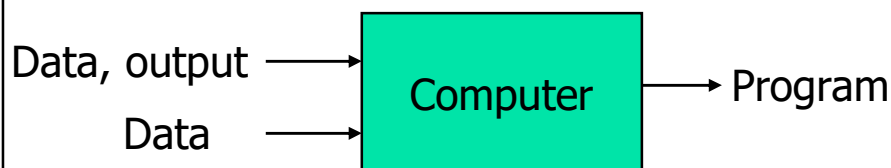
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Types of Learning

■ Semi-supervised learning

- Training data includes a few desired outputs



■ Reinforcement learning

- Rewards from sequence of actions

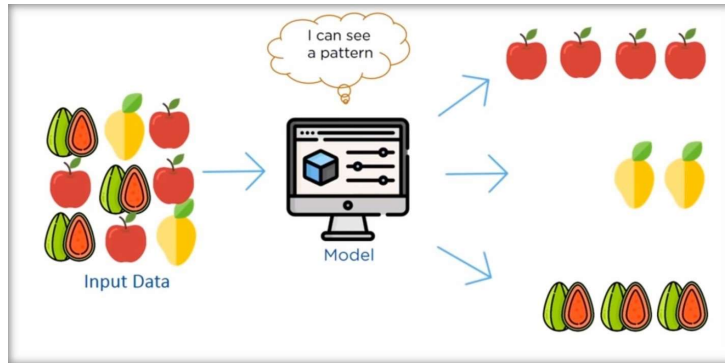
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Types of Learning



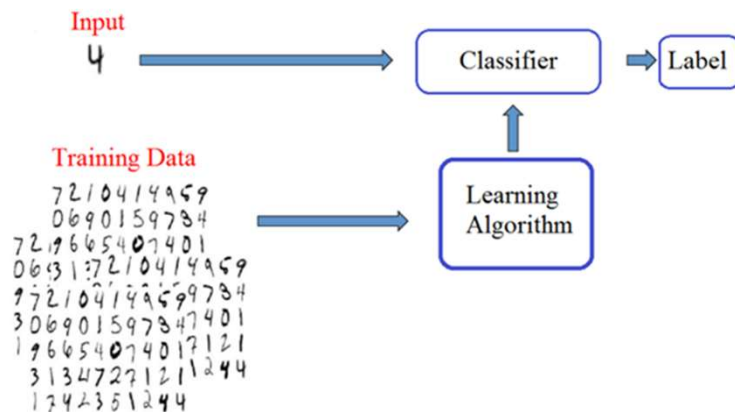
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Types of Learning



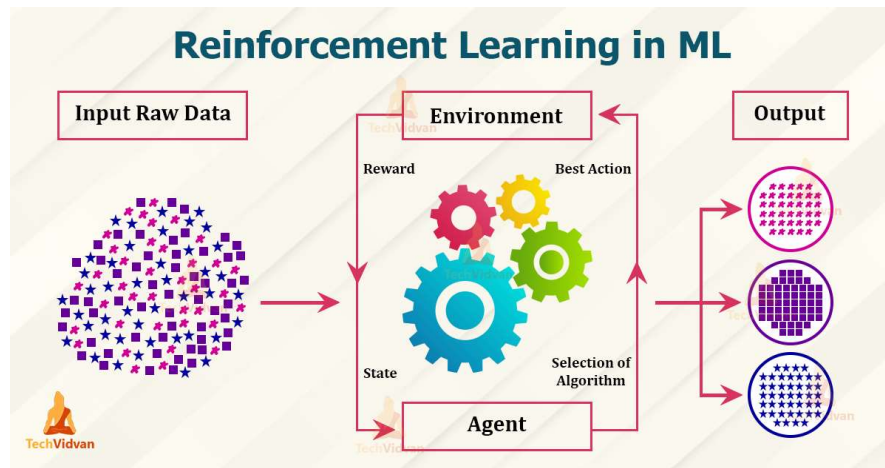
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Types of Learning



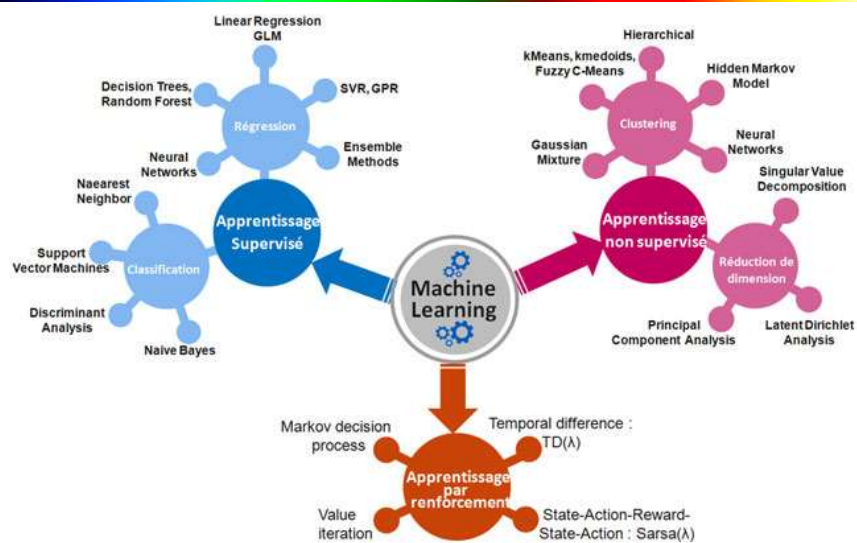
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Types of Learning



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Supervised vs. Unsupervised Learning

- **Supervised learning (classification)**
 - Supervision: The training data (observations, measurements, etc.) are accompanied by labels indicating the class of the observations
 - New data is classified based on the training set
- **Unsupervised learning (clustering)**
 - The class labels of training data is unknown
 - Given a set of measurements, observations, etc. with the aim of establishing the existence of classes or clusters in the data

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Classification vs. Prediction

- **Classification**
 - predicts categorical class labels (discrete or nominal)
 - classifies data (constructs a model) based on the training set and the values (**class labels**) in a classifying attribute and uses it in classifying new data
- **Prediction**
 - models continuous-valued functions, i.e., predicts unknown or missing values
- **Typical applications**
 - Credit/loan approval:
 - Medical diagnosis: if a tumor is cancerous or benign
 - Fraud detection: if a transaction is fraudulent
 - Web page categorization: which category it is

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Real-world machine learning use cases

- **Speech recognition:** automatic speech recognition (ASR), computer speech recognition, or speech-to-text
- **Customer service:** Online chatbots
- **Computer vision:** derive meaningful information from digital images, videos and other visual inputs
- **Recommendation engines**
- **Automated stock trading**
- **Business intelligence**
- **Self-driving cars**
- **Facial recognition**
- **Bioinformatics and Medical Diagnosis**

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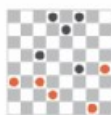
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Machine learning timeline

1943
Mathematical neural network modeling: Logician Walter Pitts and neuroscientist Warren McCulloch published the world's first mathematical modeling of a neural network to create algorithms that mimic human thought processes.



1952
Arthur Samuel's Checkers program: Arthur Samuel wrote a program for an IBM computer that had the capability to play checkers and improve each time it played.

1957
Frank Rosenblatt's perceptron: Frank Rosenblatt attempted to design the first computer neural network called "perceptron," which was built to receive visual inputs such as images, and create outputs such as labels and categorizations in response.



1963
Donald Michie's machine: Donald Michie developed a program called **MENACE** (Matchbox Educable Noughts and Crosses Engine), which had the capability to learn how to play a perfect game of tic-tac-toe (known as Noughts and Crosses in the United Kingdom).

1950
The Turing Test: Alan Turing proposed the Turing Test—according to which, if a machine can convince a human being into believing that the machine is also a human, then it is deemed "intelligent."



1956
Dartmouth Summer Research Project on Artificial Intelligence: John McCarthy invited many well-known mathematicians, researchers and scientists to **Dartmouth College**, where they brainstormed ideas on thinking machines for about six to eight weeks. This event is considered the birthplace of artificial intelligence.



1967
Nearest neighbor algorithm: The nearest neighbor algorithm granted computers the capability for basic pattern recognition, and was used to plan routes for traveling salesmen.

1970
Seppe Linnainmaa and automatic differentiation: In 1970, Finnish mathematician

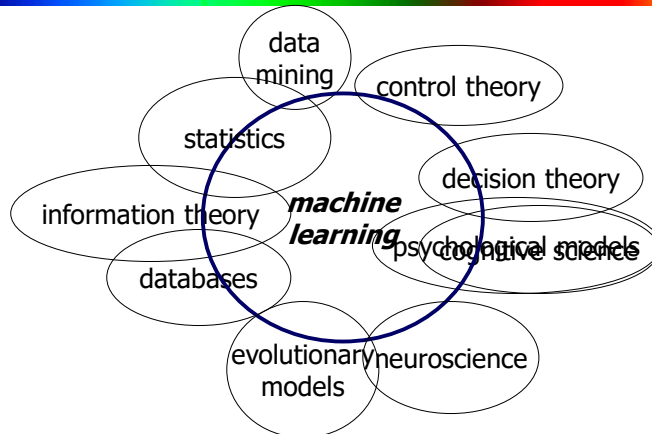
<https://searchenterpriseai.techtarget.com/definition/machine-learning/timeline>

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Related Fields



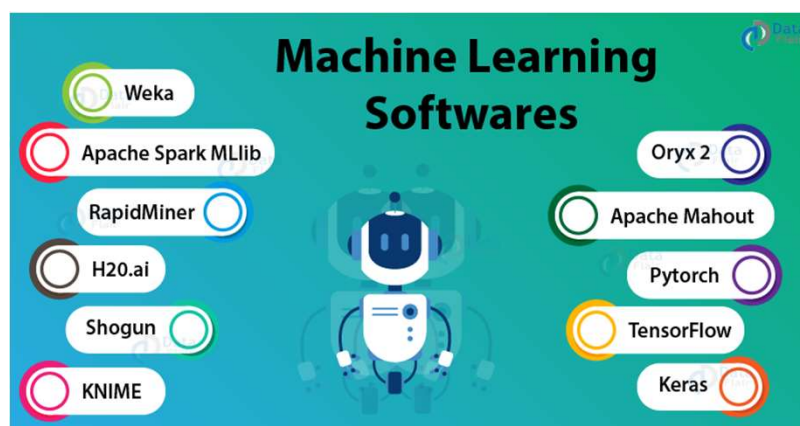
Machine learning is primarily concerned with the accuracy and effectiveness of the *computer system*.

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Machine Learning Softwares



<https://data-flair.training/blogs/machine-learning-software/>

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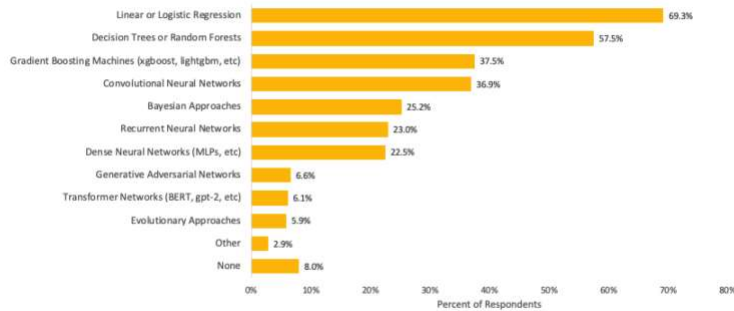
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Most used machine learning algorithms

Which of the following ML algorithms do you use on a regular basis? (Select all that apply)



Note: Data are from the 2019 Kaggle ML and Data Science Survey. You can learn more about the study here: <https://www.kaggle.com/c/kaggle-survey-2019/data>. A total of 19717 respondents completed the survey; the percentages in the graph are based on a total of 14762 respondents who provided an answer to this question.



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<https://customerthink.com/top-machine-learning-algorithms-frameworks-tools-and-products-used-by-data-scientists/>

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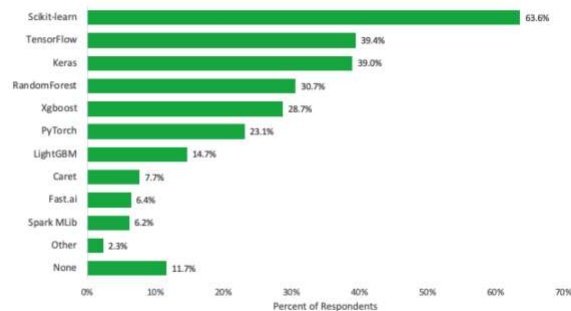
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Machine learning frameworks

Which of the following ML frameworks do you use on a regular basis? (Select all that apply)



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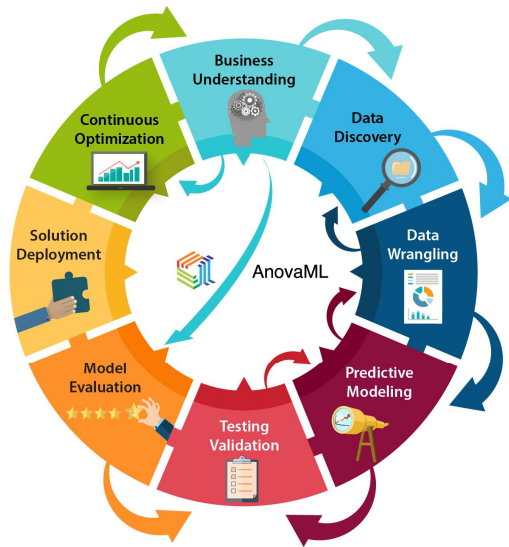
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Machine Learning Process

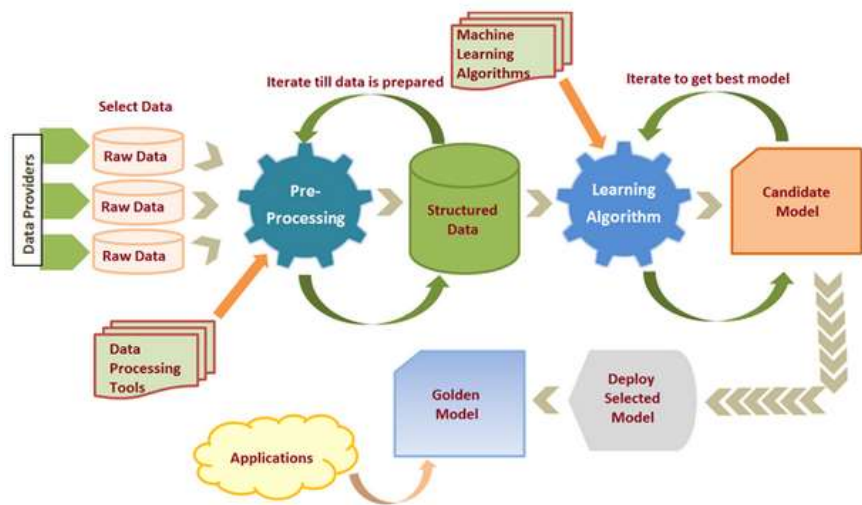


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Machine learning process



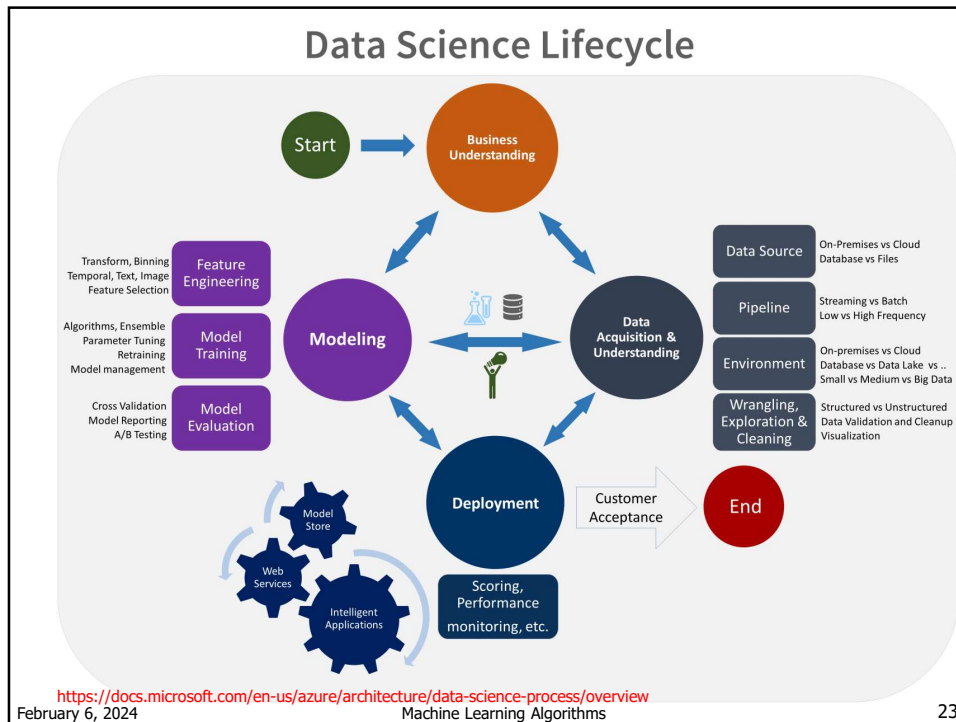
<https://www.codeproject.com/Articles/1189977/Machine-Learning-Process-and-Scenarios>

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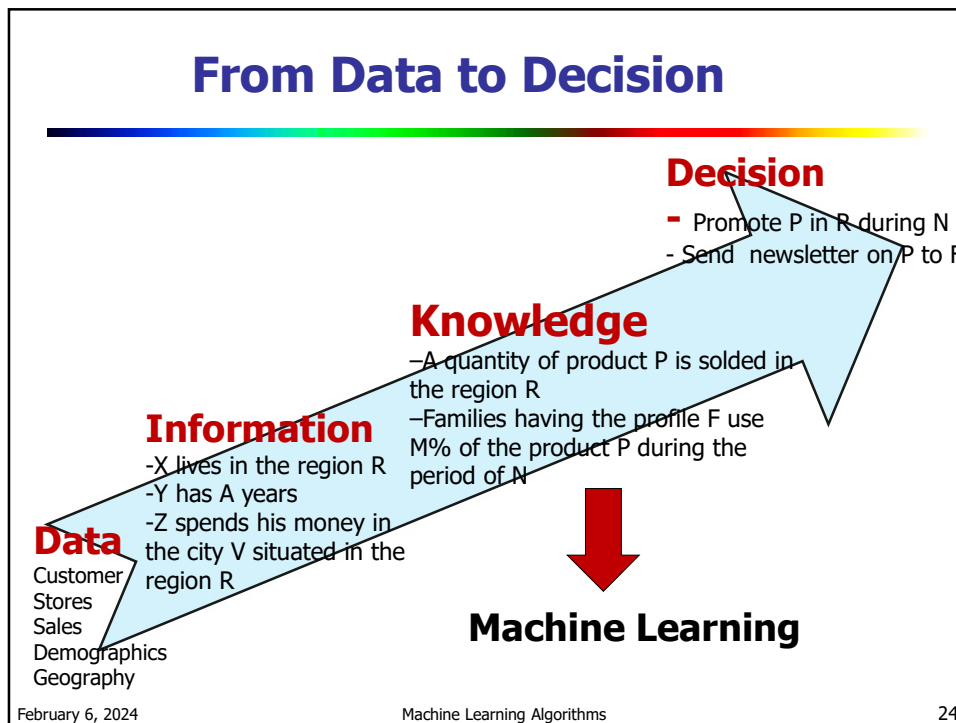
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Data characteristics : types of Data Sets

- Record

- Relational records
- Data matrix, e.g., numerical matrix, crosstabs
- Document data: text documents: term-frequency vector
- Transaction data

	team	coach	pts	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

- Graph

- World Wide Web
- Social or information networks
- Molecular Structures

TID	Items
1	Bread, Coke, Milk
2	Juice , Bread
3	Juice, Coke, Diaper, Milk
4	Juice, Bread, Diaper, Milk
5	Coke, Diaper, Milk

- Ordered

- Spatial data: maps
- Temporal data: time-series
- Sequential Data: transaction sequences
- Genetic sequence data

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Data characteristics

- Dimensionality
 - Curse of dimensionality
- Sparsity
 - Only presence counts
- Resolution
 - Patterns depend on the scale
- Similarity
 - Distance measure

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Data Characteristics :Why Is Data Dirty?

- Incomplete data may come from
 - "Not applicable" data value when collected
 - Different considerations between the time when the data was collected and when it is analyzed.
 - Human/hardware/software problems
- Noisy data (incorrect values) may come from
 - Faulty data collection instruments
 - Human or computer error at data entry
 - Errors in data transmission
- Inconsistent data may come from
 - Different data sources
 - Functional dependency violation (e.g., modify some linked data)
- Duplicate records also need data cleaning

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Data characteristics: Data Transformation

- A function that maps the entire set of values of a given attribute to a new set of replacement values s.t. each old value can be identified with one of the new values
- Methods
 - Smoothing: Remove noise from data
 - Normalization: Scaled to fall within a small, specified range
 - min-max normalization
 - z-score normalization
 - normalization by decimal scaling
 - Attribute/feature construction
 - New attributes constructed from the given ones

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Data reduction :Data Reduction Strategies

- Why data reduction?
 - A database/data warehouse may store terabytes of data
 - Complex data analysis/mining may take a very long time to run on the complete data set
- Data reduction: Obtain a reduced representation of the data set that is much smaller in volume but yet produce the same (or almost the same) analytical results
 - Dimensionality reduction — e.g., remove unimportant attributes
 - Principal component analysis
 - Singular value decomposition
 - Supervised and nonlinear techniques (e.g., feature selection)
 - Numerosity reduction (some simply call it: Data Reduction)
 - Data cub aggregation
 - Data compression
 - Regression
 - Discretization (and concept hierarchy generation)

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Evaluation of Knowledge

- Are all knowledge interesting?
 - Tremendous amount of “patterns” and knowledge
 - Some may fit only certain dimension space (time, location, ...)
 - Some may not be representative, may be transient, ...
- Interesting knowledge?
 - Coverage
 - Typicality vs. novelty
 - Accuracy
 - Timeliness

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SUPERVISED MACHINE LEARNING

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Classification and Prediction



- **What is classification? What is prediction?**
- **Issues regarding classification and prediction**
- **Classification by decision tree induction**
- **Rule-based classification**

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Supervised vs. Unsupervised Learning

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Classification—A Two-Step Process

- **Model construction:** describing a set of predetermined classes
 - Each tuple/sample is assumed to belong to a predefined class, as determined by the **class label attribute**
 - The set of tuples used for model construction is **training set**
 - The model is represented as classification rules, decision trees, or mathematical formulae
- **Model usage:** for classifying future or unknown objects
 - **Estimate accuracy** of the model
 - The known label of test sample is compared with the classified result from the model
 - Accuracy rate is the percentage of test set samples that are correctly classified by the model
 - Test set is independent of training set, otherwise over-fitting will occur
 - If the accuracy is acceptable, use the model to **classify data** tuples whose class labels are not known

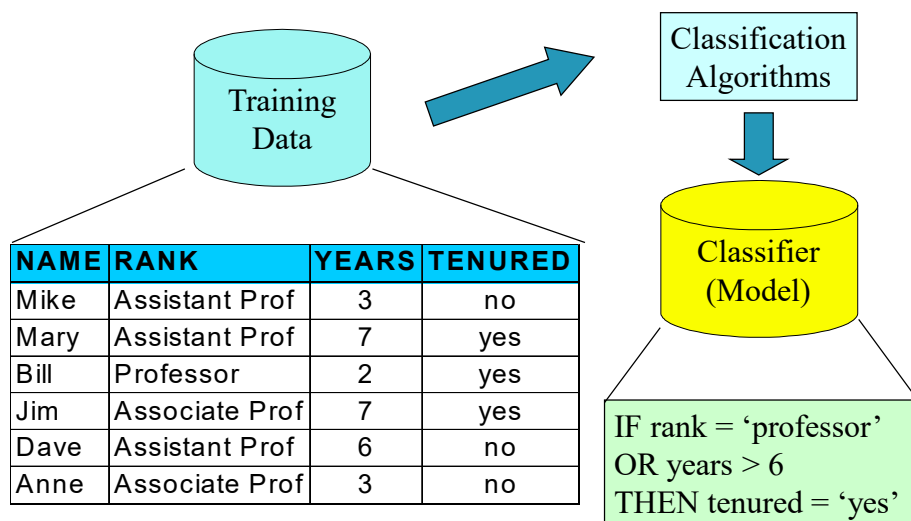
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Process (1): Model Construction



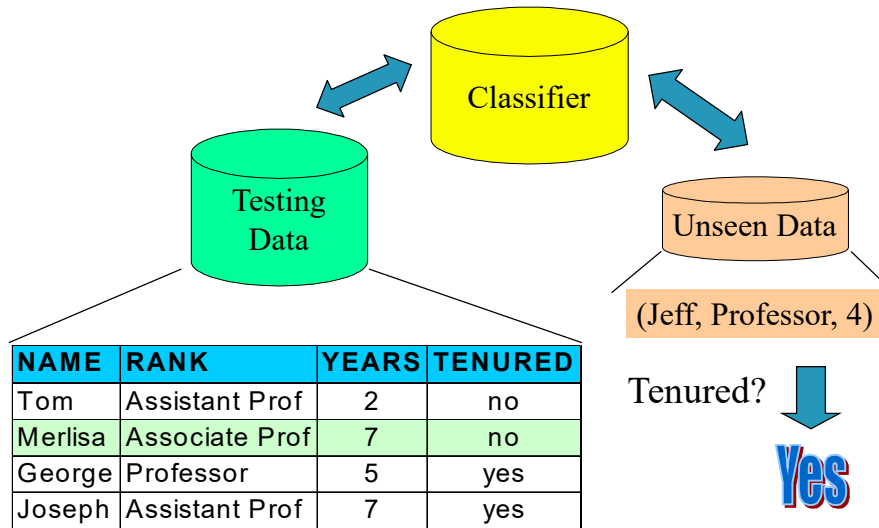
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Process (2): Using the Model in Prediction



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Test set/ training set/validation

- Holdout method
 - Given data is randomly partitioned into two independent sets
 - Training set (e.g., 2/3) for model construction
 - Test set (e.g., 1/3) for accuracy estimation
 - Random sampling: a variation of holdout
 - Repeat holdout k times, accuracy = avg. of the accuracies obtained

<https://towardsdatascience.com/cross-validation-c4fae714f1c5>

<https://www.xenonstack.com/insights/what-is-model-validation-testing>

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Test set/ training set/validation

- Cross-validation (k -fold, where $k = 10$ is most popular)
 - Randomly partition the data into k *mutually exclusive* subsets, each approximately equal size
 - At i -th iteration, use D_i as test set and others as training set
 - Leave-one-out: k folds where $k = \#$ of tuples, for small sized data
 - Stratified cross-validation: folds are stratified so that class dist. in each fold is approx. the same as that in the initial data

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Test set/ training set/validation

- Bootstrap
 - Works well with small data sets
 - Samples the given training tuples uniformly *with replacement*
 - i.e., each time a tuple is selected, it is equally likely to be selected again and re-added to the training set

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Test set/ training set/validation

- Several bootstrap methods, and a common one is **.632 bootstrap**
 - Suppose we are given a data set of d tuples. The data set is sampled n times, with replacement, resulting in a training set of n samples. The data tuples that did not make it into the training set end up forming the test set.
 - About 63.2% of the original data will end up in the bootstrap, and the remaining 36.8% will form the test set (since $(1 - 1/n)^n \approx e^{-1} = 0.368$)
 - Repeat the sampling procedure k times, overall accuracy of the model:

$$acc(M) = \sum_{i=1}^k (0.632 \times acc(M_i)_{test_set} + 0.368 \times acc(M_i)_{train_set})$$

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Test set/ training set/validation

Cross Validation

<https://towardsdatascience.com/cross-validation-c4fae714f1c5>



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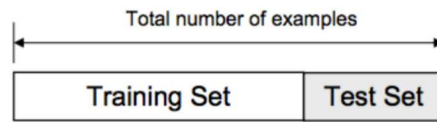
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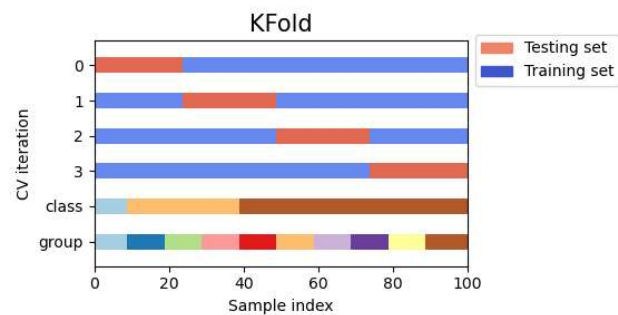
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Test set/ training set/validation

Hold-out based Validation



K-Fold Cross Validation



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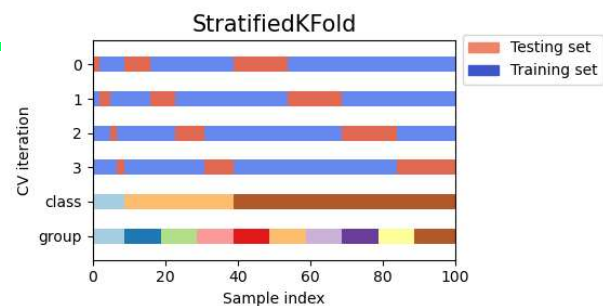
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Test set/ training set/validation

Stratified kFold



Leave-One-Out Cross Validation

special case of
k-fold cross
validation
where $k = n$ and



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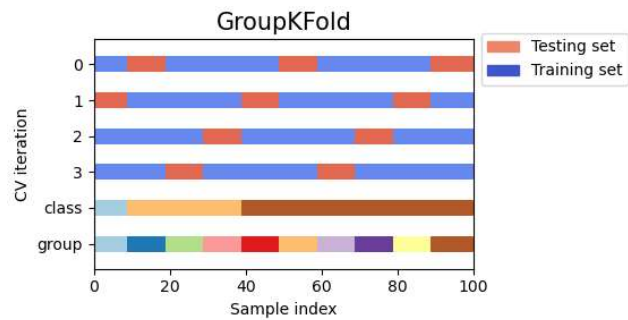
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Test set/ training set/validation

Group K-fold Cross Validation

Such a grouping of data is domain specific. An example would be when there is medical data collected from multiple patients, with multiple samples taken from each patient



DECISION TREE

Decision Tree Induction: Training Dataset

This follows an example of Quinlan's ID3 (Playing Tennis)

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

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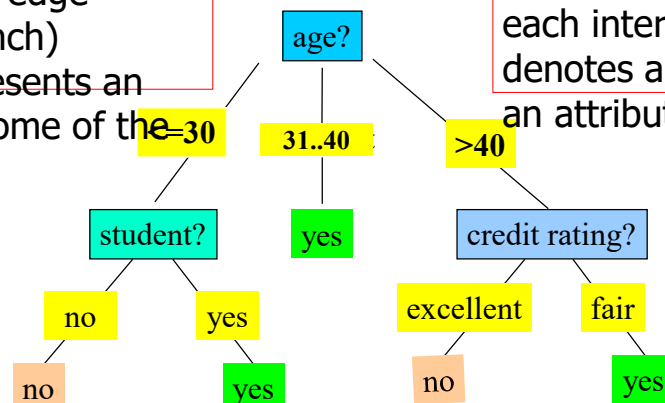
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Output: A Decision Tree for "buys_computer"

each edge (branch) represents an outcome of the test

each internal node denotes a test on an attribute

each leaf node models a class

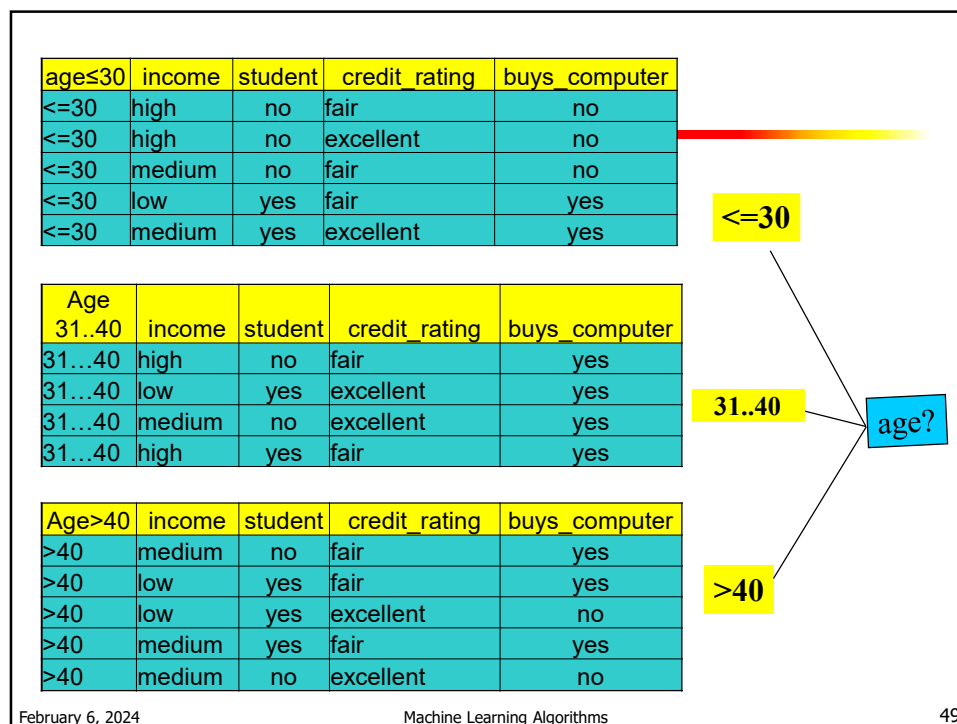


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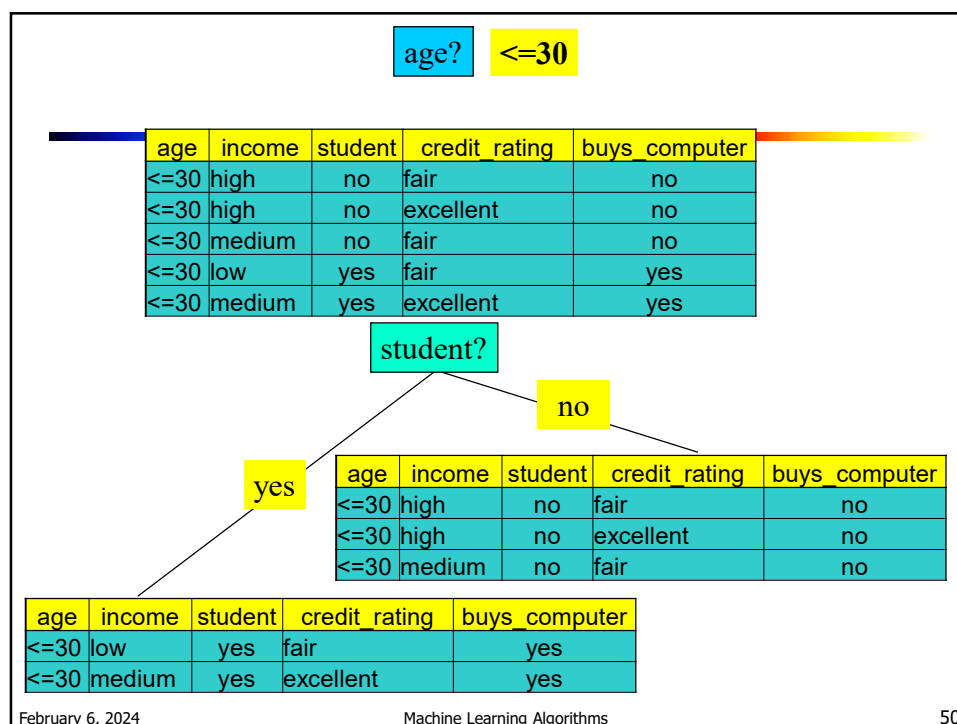
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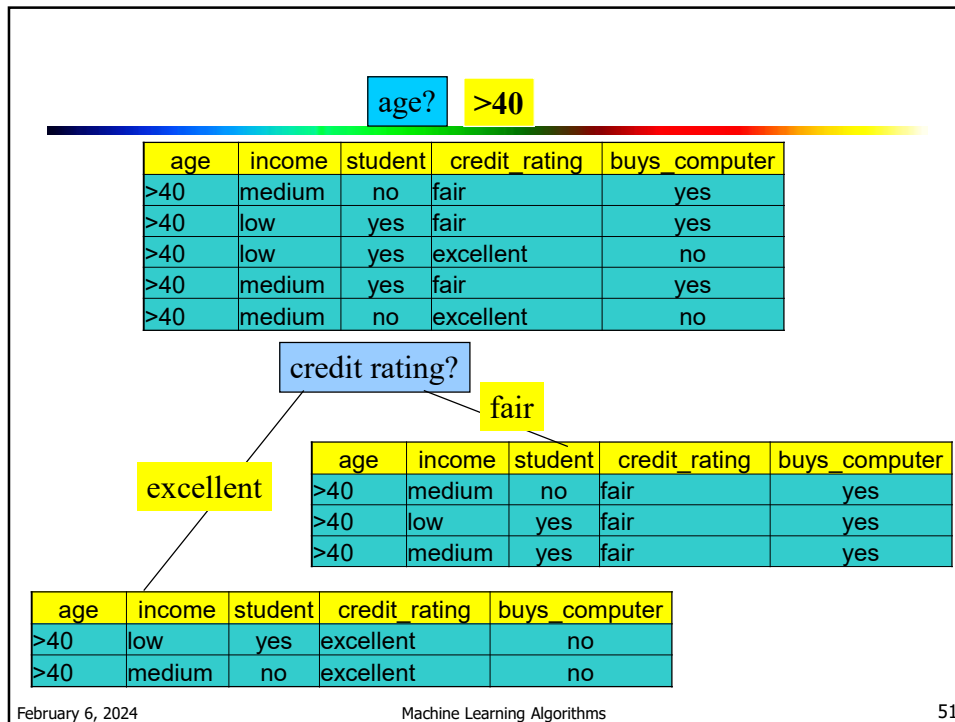
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Algorithm for Decision Tree Induction

- Basic algorithm (a greedy algorithm) (ID3, C4.5, C5, CART)
 - Tree is constructed in a **top-down recursive divide-and-conquer manner**
 - At start, all the training examples are at the root
 - Attributes are categorical (if continuous-valued, they are discretized in advance)
 - Examples are partitioned recursively based on selected attributes
 - Test attributes are selected on the basis of a heuristic or statistical measure (e.g., **information gain**)
- Conditions for stopping partitioning
 - All samples for a given node belong to the same class
 - There are no remaining attributes for further partitioning – **majority voting** is employed for classifying the leaf
 - There are no samples left

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Induction Algorithm (1)

Algorithm Generate_Decision_Tree (D, A, S) Generate a decision tree from the training tuples of data partition D

Input:

- Data partition, D, which is a set of training tuples and their associated class labels
- *Attribute_list*, A, the set of candidate attributes
- *Attribute_selection_method*, S, a method for selecting the splitting criterion that "best" partitions the data tuples into individual classes. The criterion consists of a *splitting_attribute* and, possibly, either a split point or splitting subset.

Output:

- a decision tree

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Induction Algorithm (2)

BEGIN

- (1) Create a node N;
- (2) **If** tuples in D are all of the same class, C, **then**
- (3) **return** N as a leaf node labeled with the class C
- (4) **If** *attribute_list* is empty ($A=\emptyset$), **then**
- (5) **return** N as a leaf node labeled with the majority class in D (majority voting);
- (6) Apply ***Attribute_selection_method(D, attribute_list)*** to **find** the "best" *splitting_attribute* a_{best} with its *splitting_criterion* c_{best}
- (7) Label node N with *splitting_attribute* a_{best} with its *splitting_criterion* c_{best}

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Induction Algorithm (3)

- (8) **If** splitting_attribute a_{best} is discrete-valued and multiway splits allowed **then**
- (9) Let $A \leftarrow A \setminus \{a_{\text{best}}\}$
- (10) **for each** test outcome t_j of criteria c_{best}
- (11) compute D_j the subset of tuples satisfying outcome t_j
- (12) **If** ($D_j = \emptyset$) **then**
- (13) attach a leaf node to N labeled with the majority class (or class distribution) in D_j ;
- else**
- (14) attach the node returned by **Generate_decision_Tree**(D_j, A, S)
- End for**
- return (N)**
- END.**

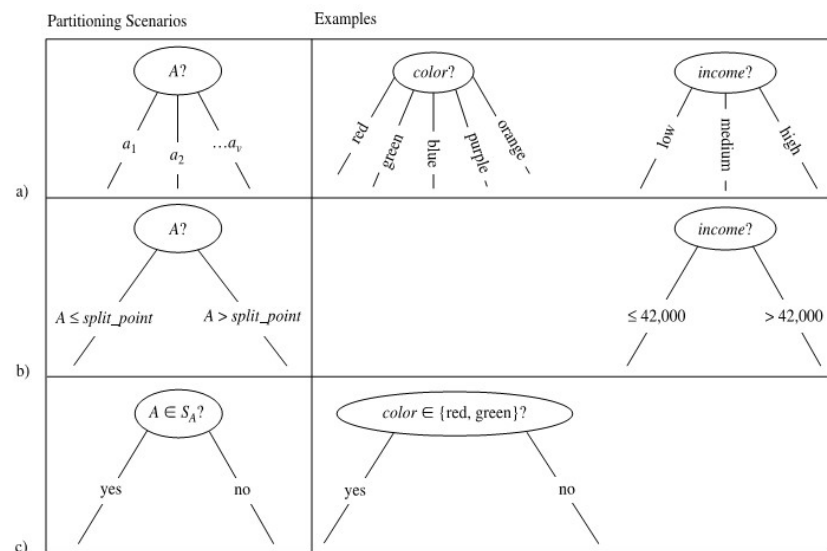
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Induction Algorithm (4)



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Induction Algorithm (5)

- The attribute selection method is used to determine the splitting attribute along with the splitting test
- Intuitively, the best splitting attribute (and criteria) is that producing partitions at each branch of the tree that are as “pure” as possible
- a partition is pure if all of its tuples belong to the same class

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Induction Algorithm (5)

- It is an heuristic for selecting attribute that “best” separate pure partitions)
- The most used heuristics are:
 - Information Gain (ID3)?
 - Gain Ratio (C4.5)
 - Gini Index (CART)

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Attribute Selection Measure: Information Gain (ID3/C4.5)

- Select the attribute with the highest information gain
- Let p_i be the probability that an arbitrary tuple in D belongs to class C_i , estimated by $|C_{i,D}|/|D|$
- **Expected information** (entropy) needed to classify a tuple in D :

$$Info(D) = - \sum_{i=1}^m p_i \log_2(p_i)$$

- **Information** needed (after using A to split D into v partitions) to classify D :

$$Info_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times I(D_j)$$

- **Information gained** by branching on attribute A

$$Gain(A) = Info(D) - Info_A(D)$$

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Attribute Selection: Information Gain

- Class P: buys_computer = "yes"
- Class N: buys_computer = "no"

$$Info_{age}(D) = \frac{5}{14} I(2,3) + \frac{4}{14} I(4,0)$$

$$Info(D) = I(9,5) = -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) = 0.940$$

$$+ \frac{5}{14} I(3,2) = 0.694$$

age	p_i	n_i	$I(p_i, n_i)$
≤ 30	2	3	0.971
31...40	4	0	0
> 40	3	2	0.971

$\frac{5}{14} I(2,3)$ means "age ≤ 30 " has 5 out of 14 samples, with 2 yes'es and 3 no's. Hence

$$Gain(age) = Info(D) - Info_{age}(D) = 0.246$$

Similarly,

$$Gain(income) = 0.029$$

$$Gain(student) = 0.151$$

$$Gain(credit_rating) = 0.048$$

age	income	student	credit_rating	buys_computer
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31...40	high	no	fair	yes
> 40	medium	no	fair	yes
> 40	low	yes	fair	yes
> 40	low	yes	excellent	no
31...40	low	yes	excellent	yes
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
> 40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
> 40	medium	no	excellent	no

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Computing Information-Gain for Continuous-Value Attributes

- Let attribute A be a continuous-valued attribute
- Must determine the *best split point* for A
 - Sort the value A in increasing order
 - Typically, the midpoint between each pair of adjacent values is considered as a possible *split point*
 - $(a_i + a_{i+1})/2$ is the midpoint between the values of a_i and a_{i+1}
 - The point with the *minimum expected information requirement* for A is selected as the split-point for A
- Split:
 - D1 is the set of tuples in D satisfying $A \leq \text{split-point}$, and D2 is the set of tuples in D satisfying $A > \text{split-point}$

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Gain Ratio for Attribute Selection (C4.5)

- Information gain measure is biased towards attributes with a large number of values
- C4.5 (a successor of ID3) uses gain ratio to overcome the problem (normalization to information gain)

$$\text{SplitInfo}_A(D) = - \sum_{j=1}^v \frac{|D_j|}{|D|} \times \log_2 \left(\frac{|D_j|}{|D|} \right)$$

- $\text{GainRatio}(A) = \text{Gain}(A) / \text{SplitInfo}(A)$
- Ex. $\text{SplitInfo}_A(D) = -\frac{4}{14} \times \log_2 \left(\frac{4}{14} \right) - \frac{6}{14} \times \log_2 \left(\frac{6}{14} \right) - \frac{4}{14} \times \log_2 \left(\frac{4}{14} \right) = 0.926$
 - $\text{gain_ratio}(\text{income}) = 0.029 / 0.926 = 0.031$
- The attribute with the maximum gain ratio is selected as the splitting attribute

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Gini index (CART, IBM IntelligentMiner)

- If a data set D contains examples from n classes, gini index, $gini(D)$ is defined as

$$gini(D) = 1 - \sum_{j=1}^n p_j^2$$

where p_j is the relative frequency of class j in D

- If a data set D is split on A into two subsets D_1 and D_2 , the $gini$ index $gini_A(D)$ is defined as

$$gini_A(D) = \frac{|D_1|}{|D|} gini(D_1) + \frac{|D_2|}{|D|} gini(D_2)$$

- Reduction in Impurity:

$$\Delta gini(A) = gini(D) - gini_A(D)$$

- The attribute provides the smallest $gini_{split}(D)$ (or the largest reduction in impurity) is chosen to split the node (*need to enumerate all the possible splitting points for each attribute*)

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Gini index (CART, IBM IntelligentMiner)

- Ex. D has 9 tuples in `buys_computer = "yes"` and 5 in "no"

$$gini(D) = 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 = 0.459$$

- Suppose the attribute `income` partitions D into 10 in D_1 : {low, medium} and 4 in D_2

$$\begin{aligned} gini_{income \in \{low, medium\}}(D) &= \left(\frac{10}{14}\right) Gini(D_1) + \left(\frac{4}{14}\right) Gini(D_2) \\ &= \frac{10}{14} \left(1 - \left(\frac{6}{10}\right)^2 - \left(\frac{4}{10}\right)^2\right) + \frac{4}{14} \left(1 - \left(\frac{1}{4}\right)^2 - \left(\frac{3}{4}\right)^2\right) \\ &= 0.450 \\ &= Gini_{income \in \{high\}}(D) \end{aligned}$$

but $gini_{\{medium, high\}}$ is 0.30 and thus the best since it is the lowest

- All attributes are assumed continuous-valued
- May need other tools, e.g., clustering, to get the possible split values
- Can be modified for categorical attributes

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Comparing Attribute Selection Measures

- The three measures, in general, return good results but
 - Information gain:
 - biased towards multivalued attributes
 - Gain ratio:
 - tends to prefer unbalanced splits in which one partition is much smaller than the others
 - Gini index:
 - biased to multivalued attributes
 - has difficulty when # of classes is large
 - tends to favor tests that result in equal-sized partitions and purity in both partitions

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Overfitting and Tree Pruning

- Overfitting: An induced tree may overfit the training data
 - Too many branches, some may reflect anomalies due to noise or outliers
 - Poor accuracy for unseen samples
- Two approaches to avoid overfitting
 - Prepruning: Halt tree construction early—do not split a node if this would result in the goodness measure falling below a threshold
 - Difficult to choose an appropriate threshold
 - Postpruning: Remove branches from a “fully grown” tree—get a sequence of progressively pruned trees
 - Use a set of data different from the training data to decide which is the “best pruned tree”

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K NEAREST NEIGHBORS ALGORITHM

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The k -Nearest Neighbor Algorithm



- All instances correspond to points in the n -D space
- The nearest neighbor are defined in terms of **distance**
 $\text{dist}(\mathbf{X}_1, \mathbf{X}_2)$
- Target function could be discrete- or real- valued
- For discrete-valued, k -NN returns the most common value among the k training examples nearest to x_q

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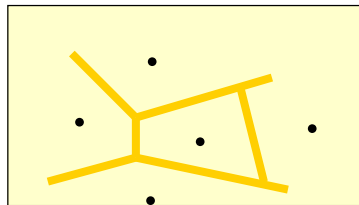
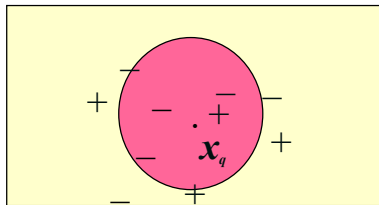
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The k -Nearest Neighbor Algorithm

- All instances correspond to points in the n -D space
- The nearest neighbor are defined in terms of Euclidean distance, $\text{dist}(\mathbf{x}_1, \mathbf{x}_2)$
- Target function could be discrete- or real- valued
- For discrete-valued, k -NN returns the most common value among the k training examples nearest to x_q
- Voronoi diagram: the decision surface induced by 1-NN for a typical set of training examples



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Discussion on the k -NN Algorithm

- k -NN for real-valued prediction for a given unknown tuple
 - Returns the mean values of the k nearest neighbors
- Distance-weighted nearest neighbor algorithm
 - Weight the contribution of each of the k neighbors according to their distance to the query x_q

$$w \equiv \frac{1}{d(x_q, x_i)^2}$$
 - Give greater weight to closer neighbors
- Robust to noisy data by averaging k -nearest neighbors
- Curse of dimensionality: distance between neighbors could be dominated by irrelevant attributes
 - To overcome it, axes stretch or elimination of the least relevant attributes

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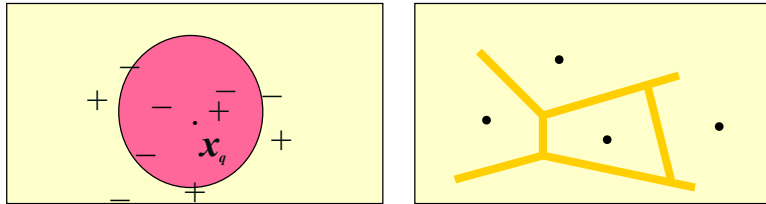
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The k -Nearest Neighbor Algorithm

- Voronoi diagram: the decision surface induced by 1-NN for a typical set of training examples



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Discussion on the k -NN Algorithm

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Discussion on the k -NN Algorithm

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- Curse of dimensionality: distance between neighbors could be dominated by irrelevant attributes
 - To overcome it, axes stretch or elimination of the least relevant attributes

SEQUENTIAL COVERING ALGORITHM

Using IF-THEN Rules for Classification

- Represent the knowledge in the form of **IF-THEN** rules
 - R: IF *age* = youth AND *student* = yes THEN *buys_computer* = yes
 - Rule antecedent/precondition vs. rule consequent
- Assessment of a rule: *coverage* and *accuracy*
 - n_{covers} = # of tuples covered by R
 - n_{correct} = # of tuples correctly classified by R
 - $\text{coverage}(R) = n_{\text{covers}} / |D|$ /* D: training data set */
 - $\text{accuracy}(R) = n_{\text{correct}} / n_{\text{covers}}$
- If more than one rule are triggered, need **conflict resolution**
 - Size ordering: assign the highest priority to the triggering rules that has the "toughest" requirement (i.e., with the *most attribute test*)
 - Class-based ordering: decreasing order of *prevalence* or *misclassification cost per class*
 - Rule-based ordering (**decision list**): rules are organized into one long priority list, according to some measure of rule quality or by experts

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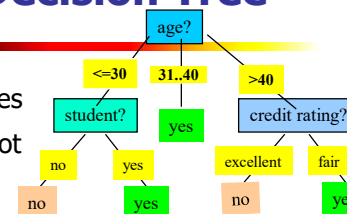
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Rule Extraction from a Decision Tree

- Rules are easier to understand than large trees
- One rule is created for each path from the root to a leaf
- Each attribute-value pair along a path forms a conjunction: the leaf holds the class prediction
- Rules are mutually exclusive and exhaustive
- Example: Rule extraction from our *buys_computer* decision-tree

IF <i>age</i> = young AND <i>student</i> = no	THEN <i>buys_computer</i> = no
IF <i>age</i> = young AND <i>student</i> = yes	THEN <i>buys_computer</i> = yes
IF <i>age</i> = mid-age	THEN <i>buys_computer</i> = yes
IF <i>age</i> = old AND <i>credit_rating</i> = excellent	THEN <i>buys_computer</i> = yes
IF <i>age</i> = young AND <i>credit_rating</i> = fair	THEN <i>buys_computer</i> = no



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Rule Induction: Sequential Covering Method

- Sequential covering algorithm: Extracts rules directly from training data
- Typical sequential covering algorithms: FOIL, AQ, CN2, RIPPER
- Rules are learned *sequentially*, each for a given class C_i will cover many tuples of C_i but none (or few) of the tuples of other classes
- Steps:
 - Rules are learned one at a time
 - Each time a rule is learned, the tuples covered by the rules are removed
 - The process repeats on the remaining tuples unless *termination condition*, e.g., when no more training examples or when the quality of a rule returned is below a user-specified threshold
- Comp. w. decision-tree induction: learning a set of rules *simultaneously*

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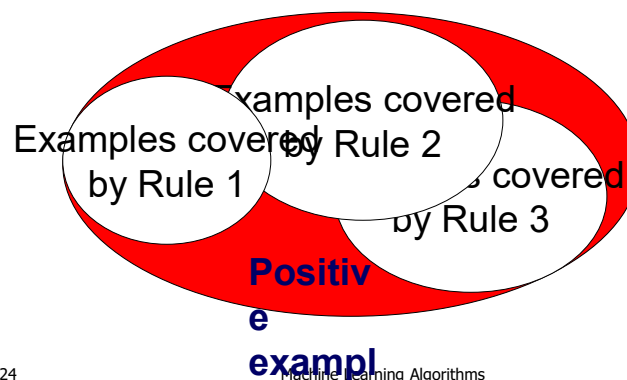
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Sequential Covering Algorithm

while (enough target tuples left)
 generate a rule
 remove positive target tuples satisfying this rule



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How to Learn-One-Rule?

- Star with the most general rule possible: condition = empty
- Adding new attributes by adopting a greedy depth-first strategy
 - Picks the one that most improves the rule quality
- Rule-Quality measures: consider both coverage and accuracy
 - Foil-gain (in FOIL & RIPPER): assesses info_gain by extending condition
- Rule pruning based on an independent set of test tuples

$$FOIL_Gain = pos' \times (\log_2 \frac{pos'}{pos' + neg'} - \log_2 \frac{pos}{pos + neg})$$

It favors rules that have high accuracy and cover many positive tuples

$$FOIL_Prune(R) = \frac{pos - neg}{pos + neg}$$

Pos/neg are # of positive/negative tuples covered by R.

If $FOIL_Prune$ is higher for the pruned version of R, prune R

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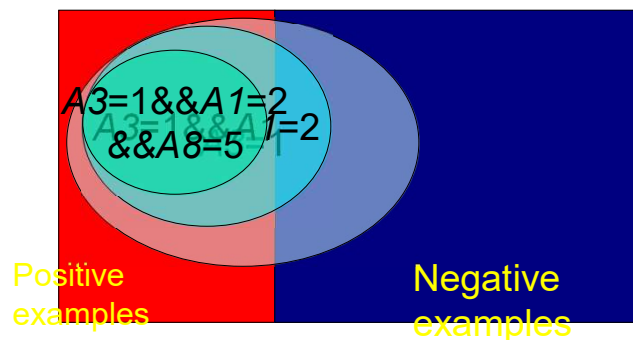
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Rule Generation

- To generate a rule
 - while(true)**
 - find the best predicate p
 - if** foil-gain(p) > threshold **then** add p to current rule
 - else break**



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SUPERVISED MACHINE LEARNING VALIDATION

Classifier Accuracy Measures

Real class\Predicted class	C_1	$\sim C_1$
C_1	True positive	False negative
$\sim C_1$	False positive	True negative

Real class\Predicted class	buy_computer = yes	buy_computer = no	total	recognition(%)
buy_computer = yes	6954	46	7000	99.34
buy_computer = no	412	2588	3000	86.27
total	7366	2634	10000	95.52

- Accuracy of a classifier M, $\text{acc}(M)$: percentage of test set tuples that are correctly classified by the model M
 - Error rate (misclassification rate) of M = $1 - \text{acc}(M)$
 - Given m classes, CM_{ij} , an entry in a **confusion matrix**, indicates # of tuples in class i that are labeled by the classifier as class j
- Alternative accuracy measures (e.g., for cancer diagnosis)
 - sensitivity = $\text{t-pos}/\text{pos}$ /* true positive recognition rate */
 - specificity = $\text{t-neg}/\text{neg}$ /* true negative recognition rate */
 - precision = $\text{t-pos}/(\text{t-pos} + \text{f-pos})$
 - accuracy = $\text{sensitivity} * \text{pos}/(\text{pos} + \text{neg}) + \text{specificity} * \text{neg}/(\text{pos} + \text{neg})$
 - This model can also be used for cost-benefit analysis

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Predictor Error Measures

- Measure predictor accuracy: measure how far off the predicted value is from the actual known value
 - Loss function**: measures the error betw. y_i and the predicted value y_i'
 - Absolute error: $|y_i - y_i'|$
 - Squared error: $(y_i - y_i')^2$
 - Test error (generalization error): the average loss over the test set
 - Mean absolute error: $\frac{\sum_{i=1}^d |y_i - y_i'|}{d}$ Mean squared error: $\frac{\sum_{i=1}^d (y_i - y_i')^2}{d}$
 - Relative absolute error: $\frac{\sum_{i=1}^d |y_i - y_i'|}{\sum_{i=1}^d |y_i - \bar{y}|}$ Relative squared error: $\frac{\sum_{i=1}^d (y_i - y_i')^2}{\sum_{i=1}^d (y_i - \bar{y})^2}$
- The mean squared-error exaggerates the presence of outliers
 Popularly use (square) root mean-square error, similarly, root relative squared error

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Error Analysis

Real class\Predicted class	C ₁	C ₂
C ₁	True positive	False negative
C ₂	False positive	True negative

Given m classes, $CM_{i,j}$ an entry in a **confusion matrix**, indicates # of tuples in class i that are labeled by the classifier as class j

Real class\Predicted class	buy_computer = yes	buy_computer = no	total	recognition(%)
buy_computer = yes	6954	46	7000	99.34
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Error Analysis

- Alternative accuracy measures (e.g., for cancer diagnosis)
 - sensitivity = $t\text{-pos}/\text{pos}$ /* true positive recognition rate */
 - specificity = $t\text{-neg}/\text{neg}$ /* true negative recognition rate */
 - precision = $t\text{-pos}/(t\text{-pos} + f\text{-pos})$
 - accuracy = $\text{sensitivity} * \text{pos}/(\text{pos} + \text{neg}) + \text{specificity} * \text{neg}/(\text{pos} + \text{neg})$
- This model can also be used for cost-benefit analysis

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 - Absolute error: $|y_i - y_i'|$
 - Squared error: $(y_i - y_i')^2$
- Test error (generalization error): the average loss over the test set
 - Mean squared error:
$$\frac{\sum_{i=1}^d (y_i - y_i')^2}{d}$$
 - Mean absolute error:
$$\frac{\sum_{i=1}^d |y_i - y_i'|}{d}$$

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Predictor Error Measures

Relative absolute error:
$$\frac{\sum_{i=1}^d |y_i - y_i'|}{\sum_{i=1}^d |y_i - \bar{y}|}$$

Relative squared error:
$$\frac{\sum_{i=1}^d (y_i - y_i')^2}{\sum_{i=1}^d (y_i - \bar{y})^2}$$

- The mean squared-error exaggerates the presence of outliers
- Popularly use (square) root mean-square error, similarly, root relative squared error

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